[INCH POUNDS]
A-A-3041A
October 21, 2003
SUPERSEDING
A-A-3041
October 24, 1996

COMMERCIAL ITEM DESCRIPTION

WRENCH, OPEND END RATCHET (TAC PATTERN) FOR TUBE FITTING, ELECTRIAL CABLE TERMINALS, AND STUFFING TUBE GLAND NUTS

The General Services Administration has authorized the use of this Commercial Item Description by all federal agencies.

1. SCOPE

1.1 SCOPE. This Commercial Item Description covers open-ratchet wrenches used for tube fitting, stuffing gland nuts and applications where difficulties arise in use of standard sockets

2. CLASSIFICATION.

2.1 Classification. This Commercial Item Description covers open ratchet, crowfoot wrench heads of the following styles and forms.

Style A - Multipurpose
Style B - Single purpose
Form A - Hexagon
Form B - Open end

3. SALIENT CHARACTERISTICS.

3.1 Design. The open end ratchet, crowfoot wrench heads shall consist of a body, slotted rotating member, an enclosure plate for accessibility in repairs, and shall be reversible by inverting the wrench head. The wrenches shall be made of forged steel of such composition as to produce wrenches conforming to the physical requirements of this Commercial Item Description. The wrench shall have a hardness of between 38 to 52 on the Rockwell C scale. The style and form of the wrenches shall be similar to Figures 1, 2, 3 and applicable. The dimensions shall be as specified in Tables I or II as applicable, except style B, form B which shall be as specified in the contract or purchase order (see 7.2).

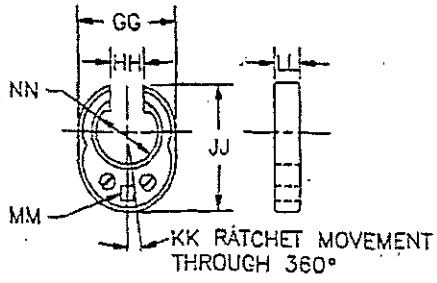


Figure 1. Style A, open ratchet, crowfoot, multipurpose wrench heads

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Hardware SuperStore (6FEE), 1500 E. Bannister Rd Bldg. #6, Kansas City, Missouri 64131.

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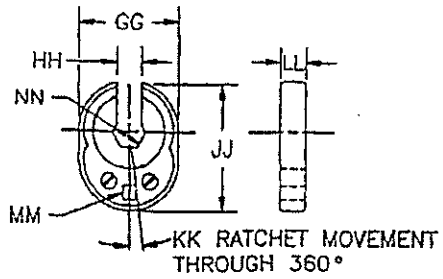


Figure 2. Style B form A, open ratchet, crowfoot, single purpose, hexagon heads.

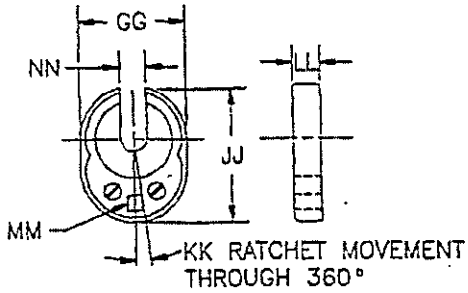


Figure 3. Style B form B, open ratchets, crowfoot, single purpose, open end wrench heads

TABLE I. Style A, open ratchet, crowfoot, multipurpose wrench heads

Component number	Rated torque (Inch-pounds)	Code	Dimensions GG (Inches)	HH (Minimum)	JJ (Minimum)	LL +.015 -.005 (Inch)	MM +.005 -.000 (Inch)	NN ± .005 (Inch)	KK Degrees	KK Minutes
2	1200	A-C	1.650	.645	2.210	.520	.381	.940	7	30
3	1500	D-F	2.150	.900	2.710	.644	.381	1.190	7	30
4	2250	G-M	2.750	1.400	3.470	.644	.508	1.720	7	30
5	3400	N-T	3.90 0	2.085	4.630	710	.508	2.470	6	0
6	3700	V-Z	4.620	2.800	5.380	.770	.508	3.190	5	0

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TABLE II. Style B, form A, open ratchet, crowfoot, single purpose, hexagon wrench heads.

Component number	Rated torque (Inch) pounds	Hex nut size (Inches)	Dimensions NN Minimum	NN Maximum	GG (Inches)	HH (Inches)	JJ (Inches)	LL +.015 -.005 (Inch)	MM +.005 -.000 (Inch)	Degrees	Minutes
10	225	9/16	.566	.575	1.630	.570 +.015 -.000	2.200	.322	.381	7	30
22	1850	1-3/4	1.759	1.775	3.467	1.796	4.167	.458	508	6	0
190	3000	2-11/16	2.701	2.722	4.645	2.706 +.020 -.000	5.367	520	508	5	0

3.2 Coatings The wrench shall be coated with black oxide or black phosphate followed by a coating of rust preventive oil, or shall be electroplated with nickel/chromium. If electroplated with nickel (Ni) coating, it shall be followed by a chromium (Cr) coating. The coating thickness requirements are as follows:

Ni = 0.000200 inch or greater
Cr = 0.000007 inch or greater

Within each area, one measurement of the thickness of each electroplated coating shall be in accordance with any of the following methods, as appropriate:

ASTM B 487: Standard Method for Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section.

ASTM B 499: Standard Test Methods for Measurements of Coating Thickness by The Magnetic Method:

ASTM B 504: Standard Method for Measurement of Thickness of Metallic Coatings by the Coulometric Method

ASTM B 530: Standard Test for Measurement of Coating Thickness by the Magnetic Method Electrodeposited Nickel Coating on Magnetic and Nonmagnetic Substrates.

ASTM B 568: Standard Test Method for Measurement of Coating Thickness by X-ray Spectrometry.

ATM B 748: Standard Method of Measurement of Thickness of Metallic Coatings by Measurement of Cross Section with a Scanning Electron Microscope.

ASTM A 754: Standard Test Method for Coating Thickness by X-ray Fluorescence.

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Visible contact marks resulting from electroplating operations shall be confined to the interior of the wrenching and drive openings. Surface of the electroplated tool shall be adherent, smooth, continuous, and free of visible defects such as blisters, pits, roughness, cracks, and uncoated areas. Surfaces shall not be stained or discolored and shall have a bright appearance. The electroplated tool shall be clean and free of damage. The tool shall be tested for adhesion of electroplated materials in accordance with ASTM B571. Standard Test Methods for Adhesion of metallic Coatings by one of the following methods: Grind-saw, Push, File, or Burnishing. The sample shall be one tool per lot.

3.3 Surface roughness. All surfaces shall be thoroughly cleaned and free from cracks, burrs, pits, nodules, apparent forge flash, and external sharp edges. All external surfaces (all surfaces excluding the wrenching surfaces and internal parts of the female drive) of the wrenches shall be fully polished to a maximum surface roughness height value of 63 micro-inches. The measurements shall be made in the surface area of the thickness of the tool. Measurements shall be made with a surface measuring instrument using a 0.03-inch roughness width cutoff. All surface roughness height values shall be rated as the arithmetic average. Nomenclature is defined in ANSI B46.1-85, Surface Texture. The sample unit shall be one tool per lot.

3.4 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains legible throughout the life of the item.

3.5 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, plating, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. REGULATORY REQUIREMENTS.

4.1 Recovered Materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch- pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product Conformance. The product provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawing, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for Inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation , the bid sample(s) shall be inspected for salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection

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.4 First article sample(s). When a first article requirement is specified in the contract, the first article samples(s) shall be inspected for all salient characteristics. A failure of any article sample to meet the salient characteristics shall be cause for rejection.

5.5 Torque test. When specified the nut engagement of the wrench shall be fully inserted onto the hexagon mandrel secured to a testing machine. Force shall be applied to a drive handle inserted into the square drive of the wrench. The specified loads specified in Table I and II shall be the load applied to the mandrel. The mandrel shall have a minimum hardness of 55 on the Rockwell C scale, be of sufficient length to fully engage the wrench, and conform to the following dimensions and tolerances. Hexagonal mandrel; 9/16 inch + 0.001 inch,- 0.003 inch across flats; 06356 inch minimum across corners. After application of the load test, any tool which exceeds the "C" dimensions of ANSI B107.17M for the nominal size more than 0.003 inch shall be rejected.

6. PACKAGING. Preservation, packaging and marking shall be as specified in the contract or order.

7. NOTES. This section contains information of a general or explanatory nature that is not mandatory.

7.1 Addresses for obtaining copies of reference documents:

ASTM B487, B499, B504, B530, B568, B571, B748, A754: American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

Federal Acquisition Regulation (FAR): Government Printing Office, Superintendent of Documents, Washington DC. 20401-9371.

Federal Specification and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents.

a. Title, number, and date of Commercial Item Description.
b. When applicable, the appropriate NSN's
c. When applicable, the specific styles and forms as specified in Commercial Item Description.
d. If bid samples are required, the number of samples required , and the subjective inspection requirements
e. If first article samples are required and the number of samples required.
f. If lo sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQCZ1.4

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

g. What preservation packing, packaging and marking of packages are required.
h. What dimensions are required for style B, form wrenches (see 3.1)

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7.3 National Stock Numbers (NSNs). The following list of NSN's corresponds to the wrench specified in this CID. This list may not be indicative of all possible NSN's associated with this CID.

NSN	SPECIFICATION COMPONENT NUMBER
5120-00-277-3661	2
5120-00-277-3662	3
5120-00-277-3663	4
5120-00-277-3664	5
5120-00-277-3665	6
5120-00-222-7975	10
5120-00-288-6502	22

MILITARY INTERESTS:
NONE: DoD has no registered interest in revisions and amendments to this Commercial Item Description until further notice.

Preparing Activity:
GSA-FSS

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